

Networking Security



Network Security & services Network Segmentation

- **Network Services**

- Active Directory
- Email (Exchange Servers)
- File servers
- Backups
- VPN

- **Zero Trust Architecture**

- Network Segmentation
- Network Architecture
- Network Segmentation
- Network Monitoring

Labs

- Configure NAT Policy (Source NAT)
- LDAP Integration (Authenticate users with AD Creds)
- Create a firewall policy

Active Directory (AD)

Active Directory (AD) is the *central identity, authentication, and authorization* platform for Windows-based enterprise environments.

- Key Capabilities

- **Centralized Authentication**

- Users log in using domain credentials validated by a Domain Controller (DC).

- **Authorization & Access Control**

- AD Groups define who can access servers, files, applications, and services.

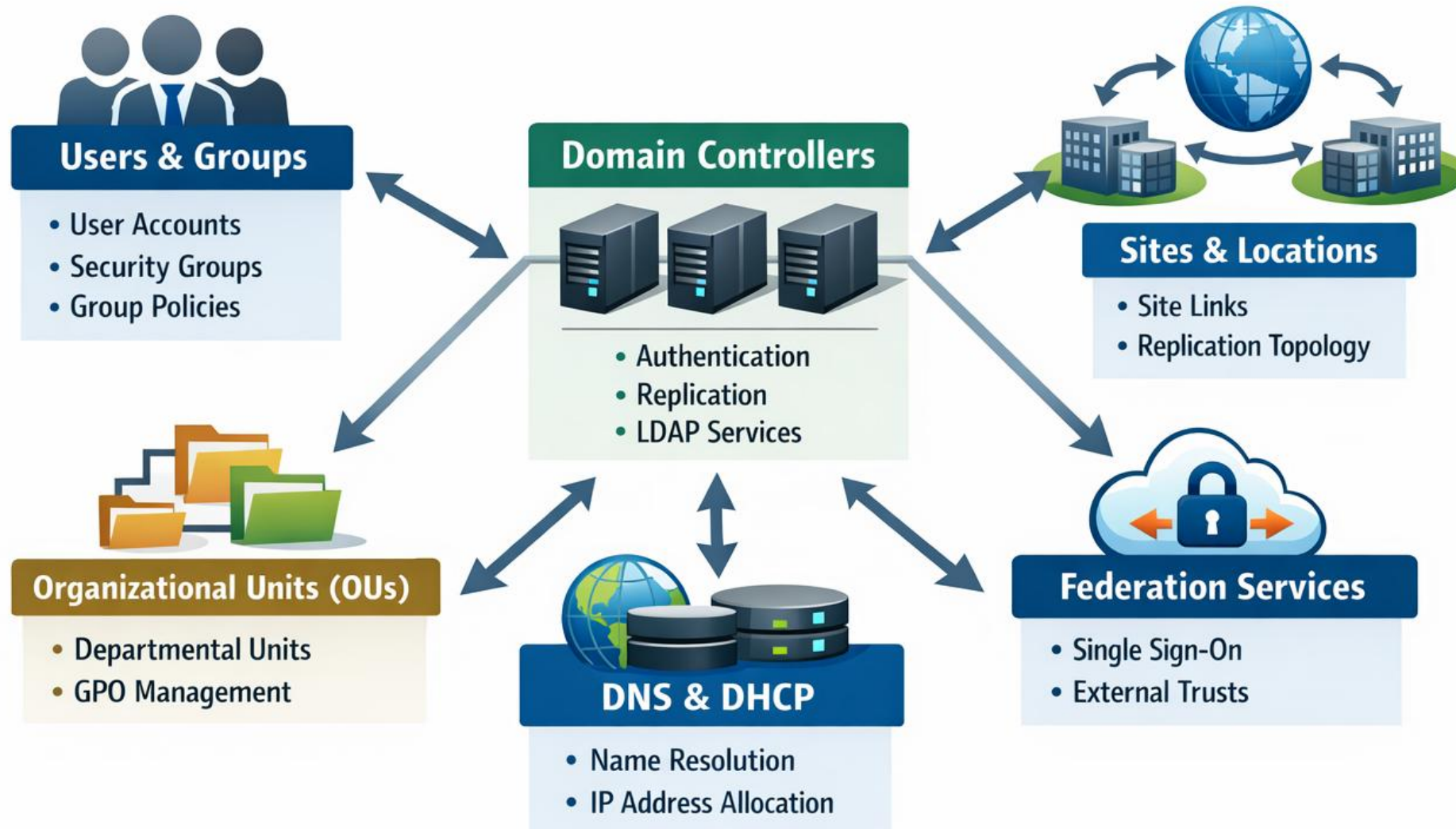
- **Policy Enforcement** (Group Policy Objects – GPOs)

- GPOs push security, system, and application policies to all domain-joined devices.

- **Directory Services**

- Stores objects: Users, Computers, Groups, Organizational Units (OUs).

Active Directory (AD) components



Active Directory

Physical Components

Data store

Domain Controller

Global Catalog Server

Read-Only Domain Controller
(RODC)

Partitions

Schema

Domain

Domain Trees

Forest

Sites

Organizational Unit (OU)

Logical Components

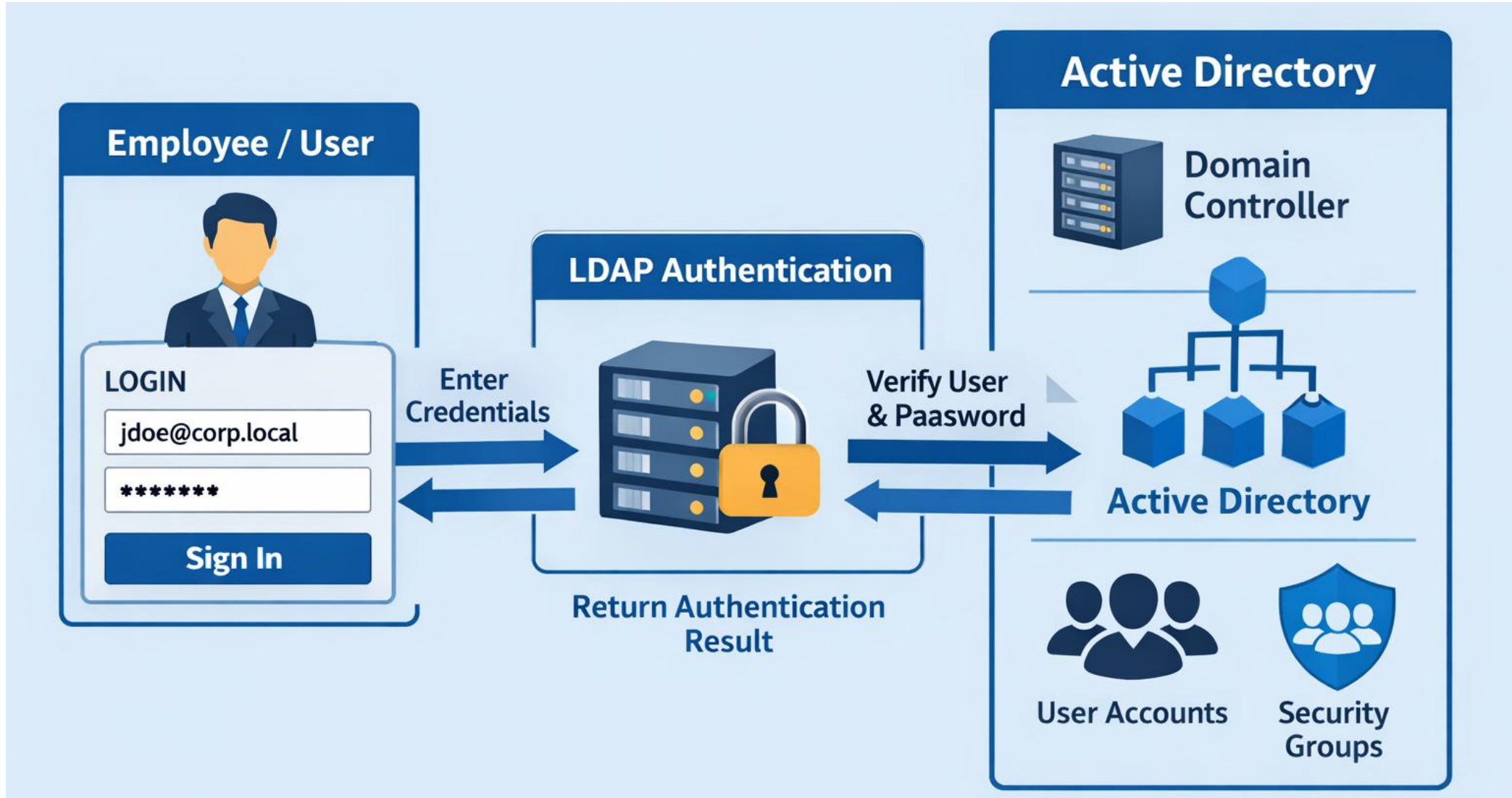


Domain Controller (DC)

- A Domain Controller is a server that is configured to store a copy of the AD DS database (NTDS.DIT) and a copy of SYSVOL folder.
- All DCs (except RODC) stores a read/write copy of NTDS.DIT and SYSVOL folder.
- Types of Domain Controller:
 - **Domain controller** – a standard domain controller.
 - **RODC** – read-only copy of DC, used in branch offices.
 - **Global Catalog (GC) server** – quick search index that stores all objects.
 - **Primary Domain Controller (PDC Emulator)** – FSMO role that manages time sync, password change and legacy compatibility.
 - **Backup Domain Controller (BDC)** – its primary backup of PDC.



Authenticating users with AD Creds



Authenticating users with AD Creds

- User identities are stored and managed in Microsoft Active Directory
- Applications and services delegate authentication to AD
- Users log in using:
 - DOMAIN\username or
 - Username@domain.com
- Passwords are never stored in the application
- AD performs:
 - Identity verification
 - Password validation
 - Account status checks (lockout, expiry, disabled)



Common Ways to Authenticate Users Using AD

- **LDAP / LDAPS Authentication (most common)**

- Used by:
 - ✓ Web applications
 - ✓ VPNs
 - ✓ Firewalls
 - ✓ Linux servers
 - ✓ Middleware applications

- **Kerberos Authentication (Domain-Joined Systems)**

- Used when:
 - ✓ Application server is domain-joined
 - ✓ Users are logged in with domain accounts

- **SAML-Based Authentication (Enterprise Web Apps)**

- Used by:
 - ✓ Web portals
 - ✓ SaaS applications
 - ✓ Identity federation

- **RADIUS Authentication (VPNs, Firewalls, Wi-Fi)**

- Used for:
 - ✓ Network access control
 - ✓ Remote VPN authentication

Email (Exchange Servers)

Microsoft Exchange provides enterprise-grade email, calendaring, and collaboration services.

- **Key Components**

- Mailbox Server Role – Stores mailbox databases.
- Client Access Services (CAS) – Provides connectivity (Outlook, OWA).
- Transport Services – Routing and delivering emails inside and outside the organization.

- **Common Integrations**

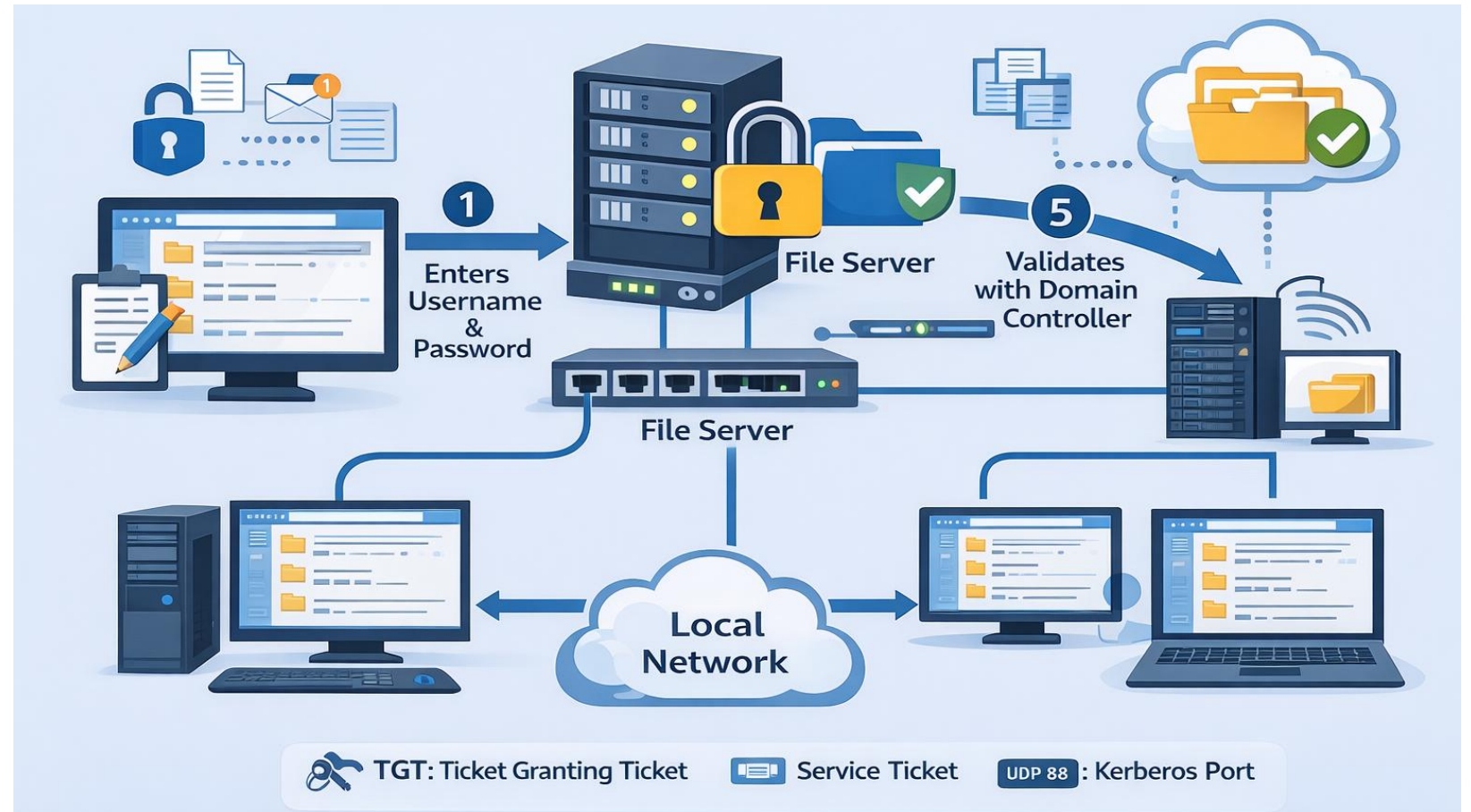
- Active Directory for authentication
- Anti-Spam & Email Security Gateways
- Mobile Device Management (EAS)

File Servers

A centralized storage solution where users store files, data, and shared information.

▪ Capabilities

- SMB file sharing
- NTFS permissions & Access Control
- Quotas & Storage policies
- DFS (Distributed File System) for replication and availability



Backups

Protect organizational data from loss, corruption, cyber-attacks (e.g., ransomware), or accidental deletion.

- **Backup Types**

- Full Backup
- Incremental Backup
- Differential Backup

- **Methods**

- On-prem backup (Veeam, Commvault, Veritas)
- Cloud-based backups (Azure Backup, AWS Backup)
- Offsite backups (for DR)



VPN (Virtual Private Network)

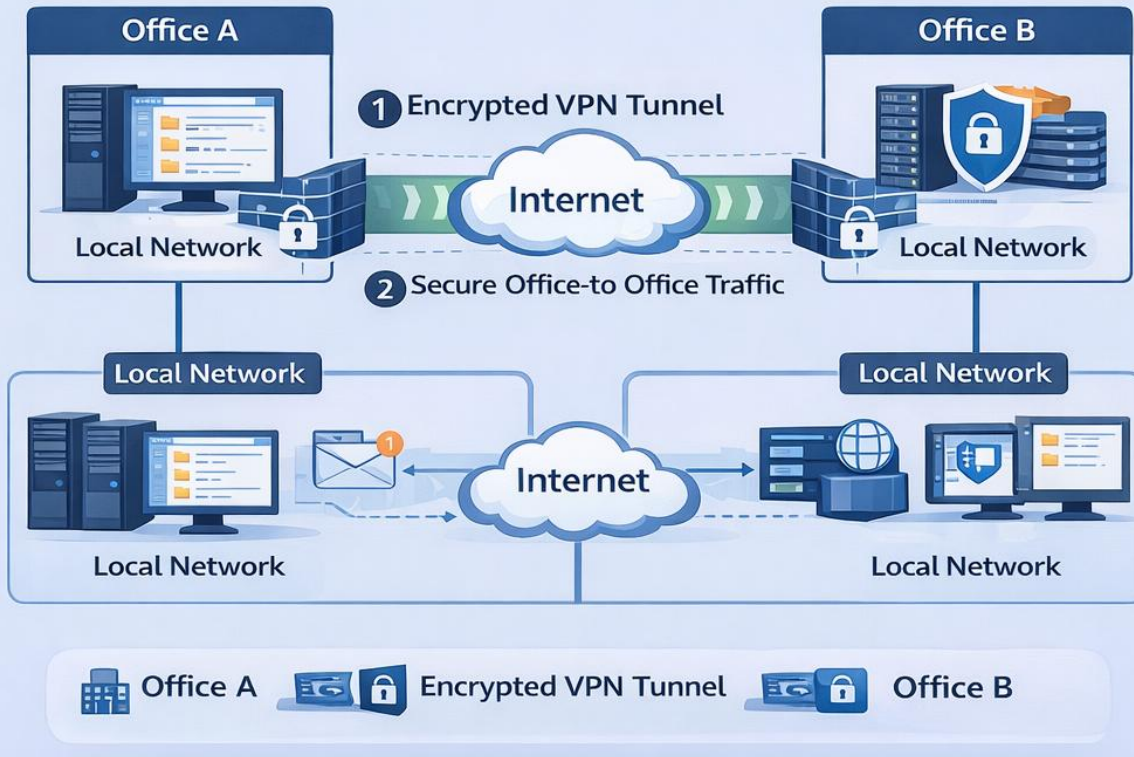
Provides secure encrypted connectivity between users or sites.

- Types of VPN
 - **Remote Access VPN**
 - Allows users to connect securely from external locations.
 - **Site-to-Site VPN**
 - Connects two offices or data centers.
- Common Technologies
 - IPsec
 - SSL VPN
 - OpenVPN
 - L2TP

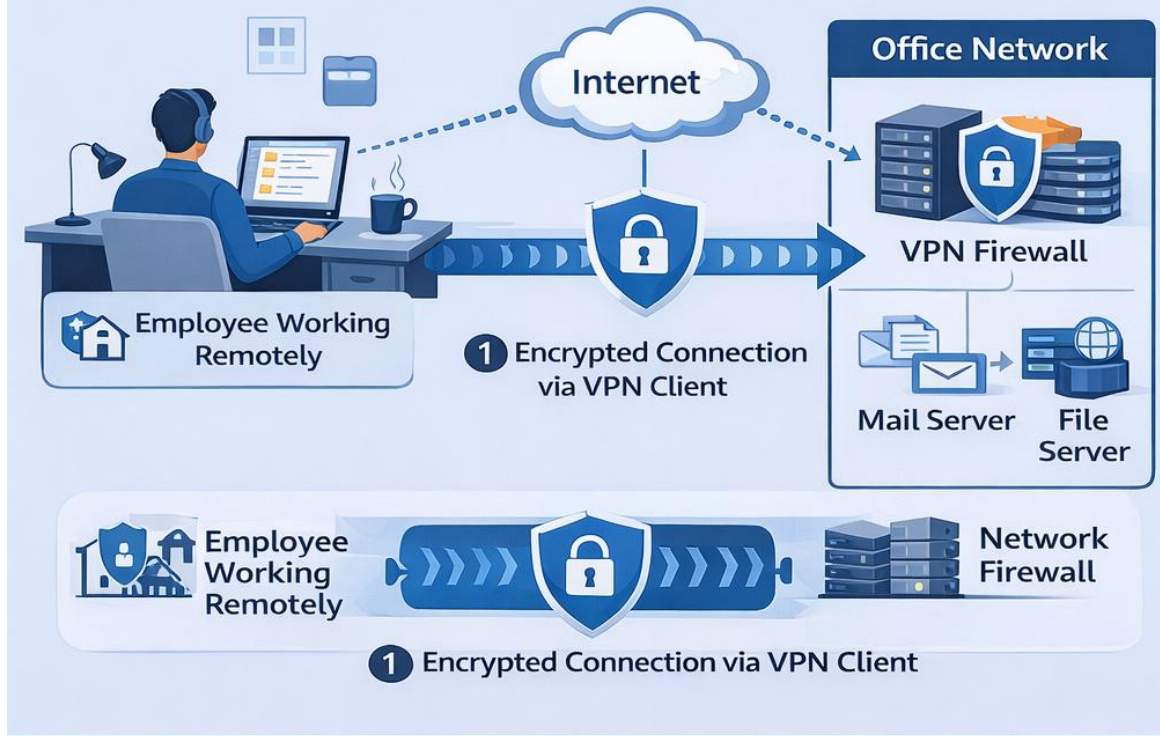


VPN (Virtual Private Network)

Basic Working of Site-to-Site VPN



Basic Working of Remote Access VPN



Zero Trust Architecture (ZTA)

Zero Trust follows the principle:

Never Trust, Always Verify.

- Every access request is validated continuously—no implicit trust exists even inside the corporate network.
- Core Pillars
 - Identity Verification (MFA, SSO, Conditional Access)
 - Device Compliance (Endpoint security, posture monitoring)
 - Least Privilege Access
 - Micro-Segmentation
 - Continuous Monitoring and Risk Evaluation



Network Segmentation

Divide the network into smaller logical segments to reduce attack spread and improve performance.

- **Types of Network Segmentation:**

- **Physical Segmentation**

- Separate network devices or cabling.

- **Logical Segmentation**

- VLANs, VRFs, Zones, Micro-segmentation.

- **Security Segmentation**

- NGFW policies separating zones like LAN, DMZ, WAN, Server Segment, User Segment.

- **Benefits**

- Limits lateral movement of attackers
 - Limits broadcast domains
 - Better performance and control
 - Enables Zero Trust



Network Architecture

The structured design of the entire network including topology, hardware, software, and policies.

▪ **Components**

- Core Layer (high-speed switching)
- Distribution Layer (routing, L3 decisions)
- Access Layer (end-user connectivity)
- DMZ (public-facing services)
- Perimeter Layer (firewalls, IPS, WAF)
- WAN & VPNs
- Wireless infrastructure

▪ **Key Goals**

- Scalability
- High-availability
- Security
- Manageability
- Performance optimization

Network Monitoring

Track network performance, availability, and security in real time.

- Tools

- Nagios / Zabbix / PRTG / SolarWinds, SNMP monitoring
- NetFlow / sFlow / IPFIX collectors
- SIEM (Splunk, QRadar, Azure Sentinel)

- Monitoring Areas

- **Performance Monitoring**

- Latency
 - Bandwidth utilization
 - Packet loss

- **Security Monitoring**

- IDS/IPS alerts
 - Firewall logs
 - VPN logs
 - User login anomalies

- **Availability Monitoring**

- Server uptime
 - Device health
 - Link failures



That's all Folks!